SPECIAL METALS

INCOLOY® alloy 25-6MO

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The nickel and chromium contents of this alloy make it resistant to a wide range of corrosive environments. The alloy is especially resistant to non-oxidizing acids such as sulfuric and phosphoric. The high molybdenum content and nitrogen provide resistance to pitting and crevice corrosion, while copper enhances resistance to sulfuric acid. The complete limiting chemical composition is given in Table 1 (UNS N08926).

INCOLOY alloy 25-6MO is a fully austenitic alloy containing 6% molybdenum and offering excellent corrosion-resistance in a wide variety of aggressive, aqueous environments. It replaces the conventional, austenitic stainless steels (AISI 316 and 317), where their capabilities are pushed to their performance limits. The alloy, therefore, falls into a "super austenitic stainless steel" category. (See the chart below.) It can also represent a cost-effective alternative to the higher nickel alloys in some marine and chemical processing environments.

One of the outstanding attributes of INCOLOY alloy 25-6MO is its resistance to environments containing chlorides or other halides. This alloy is especially suited to handle high-chloride environments such as brackish water, sea water, caustic chlorides and pulp mill bleach systems. Applications include chemical and food processing, pulp and paper bleaching plants, marine and offshore platform equipment, salt plant evaporators, air pollution control systems, and condenser tubing, service water piping, and feedwater heaters for the power industry.

Publication No. SMC-005

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Data contained in this publication are typical for the properties described, but are not suitable for specifications unless given as limiting.

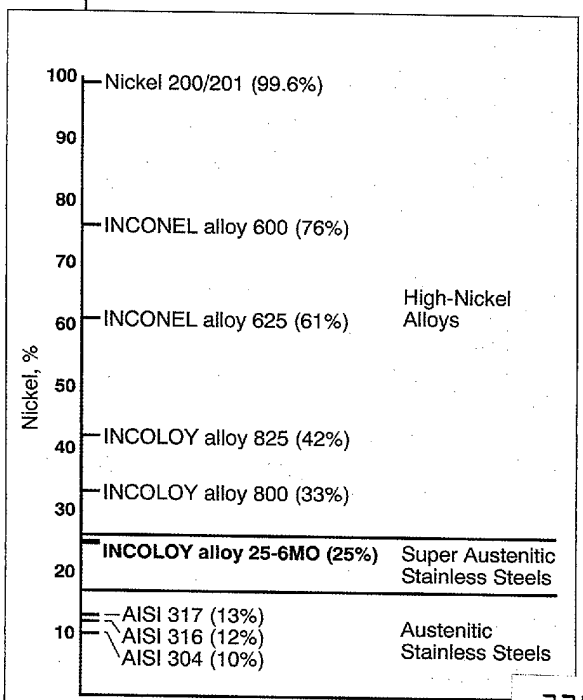


Table 1 – Limiting Chemical Composition, % (UNS N08926)

Nickel	24.00 – 26.00
Iron	Remainder
Chromium	19.00 – 21.00
Molybdenum	6.0 – 7.0
Copper	0.5 –1.5
Nitrogen	0.15 – 0.25
Carbon	0.020 max.
Manganese	2.00 max.
Phosphorus	0.03 max.
Sulfur	0.01 max.
Silicon	0.5 max.

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Pitting and Crevice Corrosion

INCOLOY alloy 25-6MO is a fully austenitic alloy offering excellent resistance to pitting and crevice corrosion. Performance in these areas is often measured using Critical Pitting Temperatures (CPT), Critical Crevice Temperatures (CCT), and Pitting Resistance Equivalent Numbers (PREN). The corrosion-resistance of stainless steels is, basically, dependent on their chemical compositions. The PREN is determined by a calculation based on the chromium, molybdenum and nitrogen contents:

PREN = % Cr + (3.3 x % Mo) + (30 x % N)

As a general rule, the higher the PREN, the better the resistance to pitting. However, alloys having similar values may differ considerably in actual service. Those with values greater than 38 on the PREN scale offer more corrosion-resistance than the austenitic stainless steels. INCOLOY alloy 25-6MO, with its PREN of 47, offers a cost-effective means of avoiding aggressive chloride attack. (See Table 2.)

The Critical Pitting Temperature (CPT) test involves exposing samples to 6% ferric chloride solutions (ASTM Standard G48[1]) and raising the temperature by incremental amounts until the onset of pitting. New, unexposed test specimens and fresh 6% $FeCl_3$ solution are used at each test temperature. The minimum accepted temperature for North Sea offshore applications is 40°C (104°F), while in pulp and paper bleaching environments, this temperature would typically be 50°C (122°F). Once again, a ranking of alloys can be achieved as shown in Table 3. The critical pitting temperature for INCOLOY alloy 25-6MO has been shown to be 65–70°C (149–158°F).

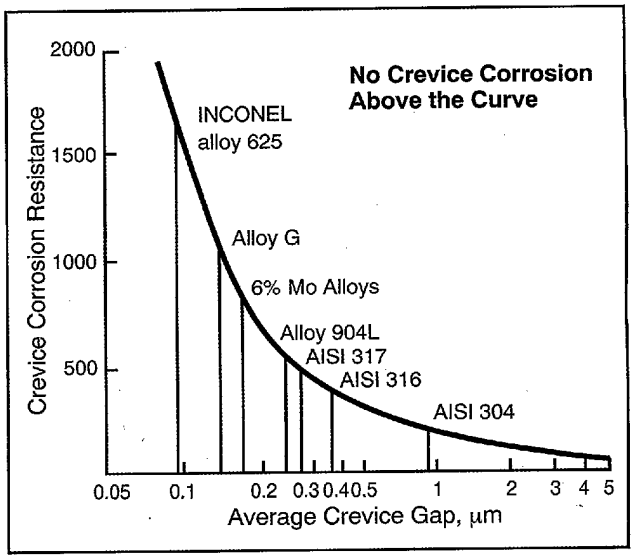
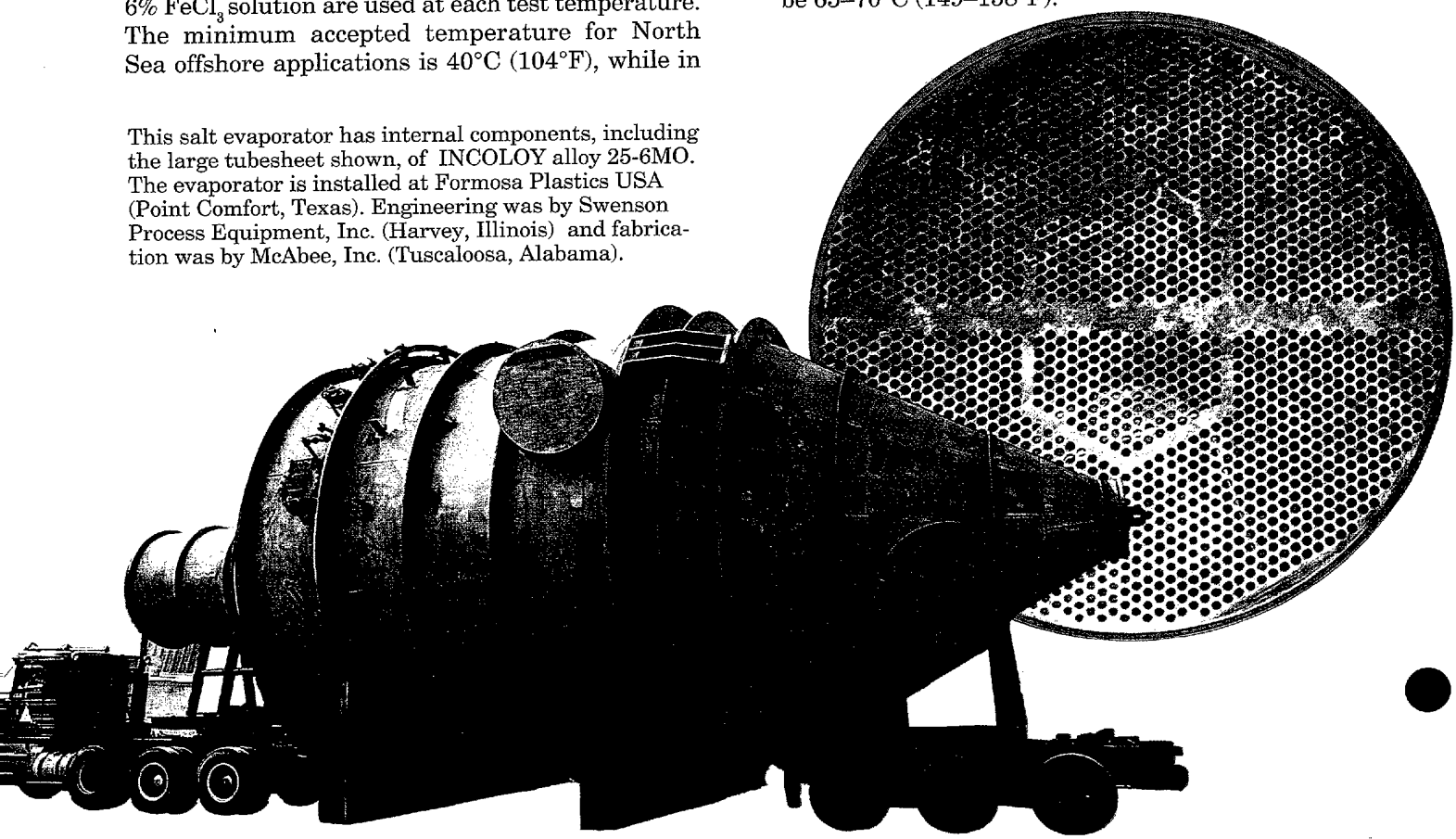


Figure 1 – Effect of crevice gap on initiation of crevice corrosion in ambient temperature sea water.

This salt evaporator has internal components, including the large tubesheet shown, of INCOLOY alloy 25-6MO. The evaporator is installed at Formosa Plastics USA (Point Comfort, Texas). Engineering was by Swenson Process Equipment, Inc. (Harvey, Illinois) and fabrication was by McAbee, Inc. (Tuscaloosa, Alabama).



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The Critical Crevice Temperature (CCT) test[2] involves exposing samples to the same aggressive ASTM G48 test solution with a multiple crevice device (TFE-fluorocarbon washer) attached to the surface of the specimen. The results are shown in Table 3 where the temperature recorded shows the onset of crevice corrosion.

A sophisticated mathematical model has been designed by Oldfield, Lee and Kain[3], showing that the crevice gap also has a major influence on the development of a critical crevice. Figure 1, based on this model, illustrates that the very resistant alloys such as INCONEL alloy 625 only corrode when extremely "tight" crevices exist. Oldfield states that the tightest gaps that are likely to be encountered in practice lie between 0.2 and 0.5 μm. Gaps of less than 0.2 μm are almost impossible to achieve and are unlikely to be encountered in service. Hence, the 6% Mo alloys offer an excellent combination of performance and economic price. Crevice corrosion will initiate on 316L stainless steel in sea water at crevice gaps of 0.4 μm. Large amounts of this alloy are used in marine service every year, but this model indicates there is a risk of crevice corrosion, and this is a fact which is borne out in practice.

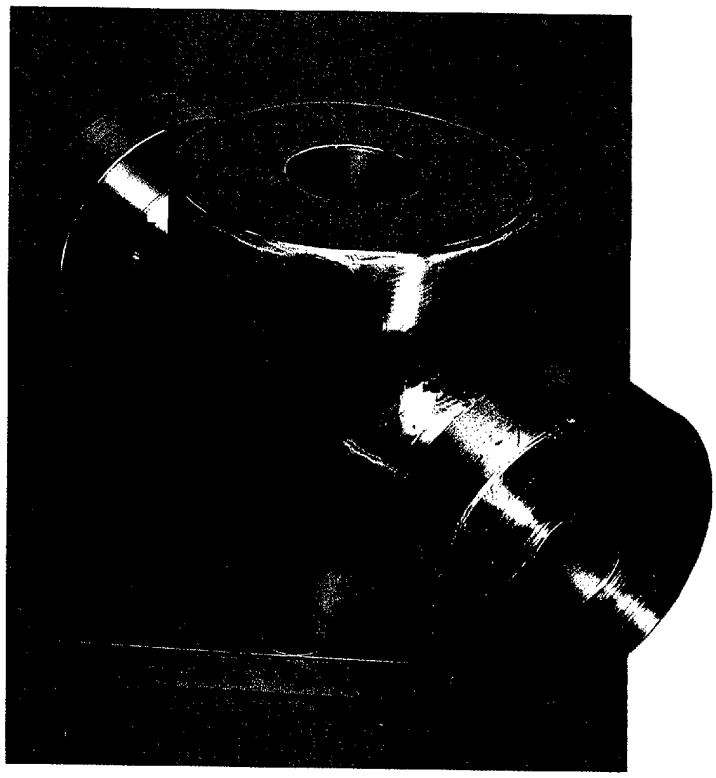
Pitting corrosion test data have also been developed for welded INCOLOY alloy 25-6MO[4]. (See Table 4.) Overmatching (molybdenum content higher than base metal) filler metal was used to ensure the integrity of the joint. INCONEL filler metal 625, with 9% molybdenum, was used.

Table 2 – Pitting Resistance Equivalent Numbers [%Cr + (3.3 x % Mo) + (30 x % N)]

Alloy	Cr, %	Mo, %	N, %	PREN
AISI 304	19	—	—	19
AISI 316	17	2.5	—	25
AISI 317	19	3.5	—	31
Alloy 904L	20	4.5	—	35
Alloy 2205	22	3.0	0.15	36
INCOLOY alloy 25-6MO	20	6.5	0.2	47

Table 3 – Critical Crevice Temperatures (CCT) and Critical Pitting Temperatures (CPT) in an Acidified 6% Ferric Chloride Solution

Alloy	CCT		CPT	
	°C	°F	°C	°F
INCONEL alloy 625	30–35	86–95	>85	>185
INCOLOY alloy 25-6MO	30–35	86–95	70	158
Alloy 904L	12.5	54	37.5	99
AISI 316L	<0	<32	20	68
AISI 317LM	20	68	52.5	126
AISI 304	<0	<32	15	59



A valve body of INCOLOY alloy 25-6MO forged by Ellwood Texas Forge (Houston, TX) from a 770 lb (349 kg) billet.

Table 4 – Results of Pitting Tests[a] on INCOLOY alloy 25-6MO Tubing Welded with INCONEL filler metal 625

Welding Method	Welding Current Level	Pitting		
		Base Metal	Heat-Affected Zone	Weld
Gas Tungsten Arc	Normal	No	No	No
Gas Tungsten Arc	High	No	No	No
Gas Tungsten Arc	Low	No	No	No
Gas Metal Arc	Normal	No	No	No

[a] Tested according to ASTM G48 with 72 h exposure at 95°F (35°C). Two specimens of each type were tested.

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Stress-Corrosion Cracking

Since the fundamental work of Copson[5], chloride stress-corrosion cracking resistance of the austenitic alloys has been known to improve with increasing nickel content. (See Figure 2.) The addition of nitrogen forms a synergistic effect with the nickel, and makes an alloy appear to have a higher nickel content than it actually does. Consequently, with 25% nickel and 0.15–0.25% nitrogen, INCOLOY alloy 25-6MO offers significantly better resistance to chloride-ion stress-corrosion cracking than lower nickel content materials such as AISI 317 stainless steel. The duplex grades have increased resistance compared to conventional austenitic grades, but the "super austenitic" grades offer even better resistance[6].

The necessary conditions for stress-corrosion cracking are stress, temperature, and corrosive media. When chlorides are present, the other two conditions could be controlled to avoid this type of attack. In many cases, stresses cannot be avoided since residual stresses from fabrication, welding, and thermal cycling will be present. Consequently, an alloy with sufficient alloying additions represents the only practical solution. For that reason, 6% molybdenum alloys are used for offshore sea water service.

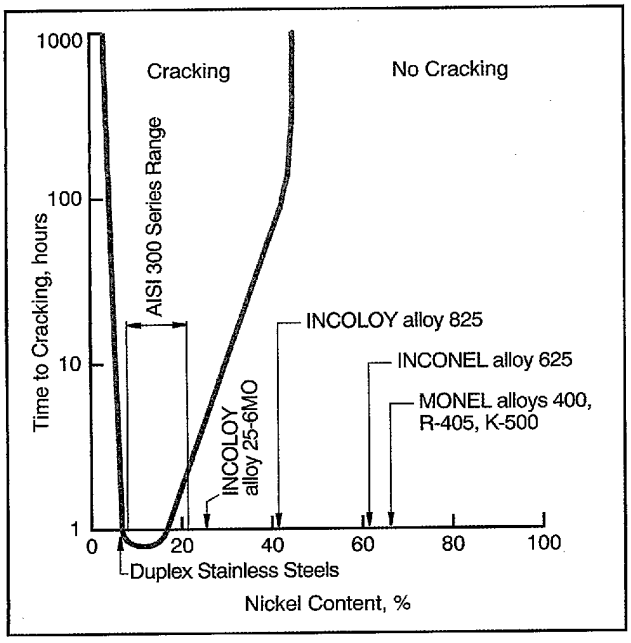


Figure 2 – The Copson U-Curve. The classic indicator of susceptibility to chloride-ion stress-corrosion cracking is the boiling 42% magnesium chloride test. INCOLOY alloy 25-6MO, with 25% nickel and an addition of nitrogen which acts synergistically with the nickel, offers significantly better resistance to chloride-ion stress-corrosion cracking than the lower nickel content stainless steels, such as AISI 317.

Intergranular Corrosion

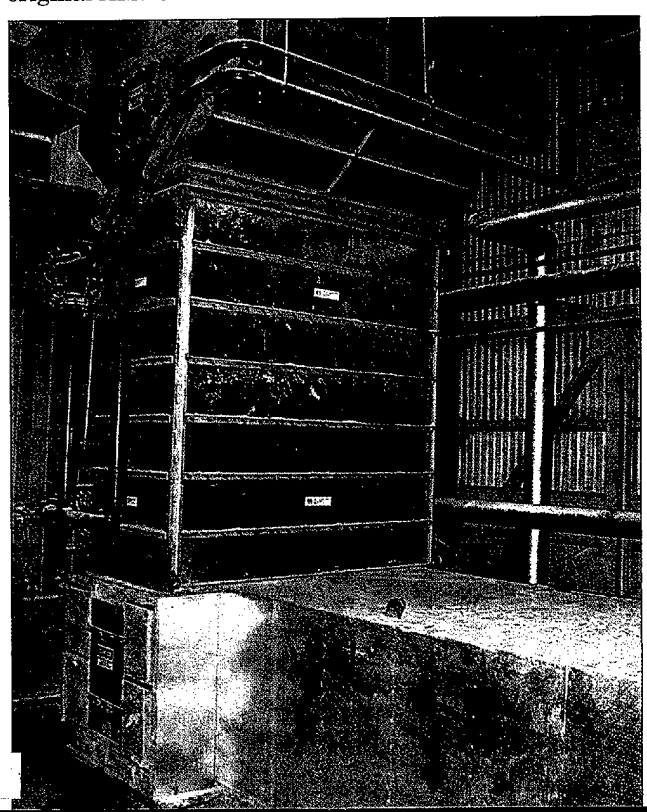
INCOLOY alloy 25-6MO has good resistance to intergranular attack. Table 5 shows the performance of the alloy in a standard test (ASTM Standard G28A[7]) for detecting susceptibility to intergranular attack. The test involves exposure for 120 h to boiling 50% sulfuric acid containing 25 g of ferric sulfate per 600 ml of acid.

Table 5 – Results of Intergranular Attack Tests[a] on INCOLOY alloy 25-6MO Sheet Specimens

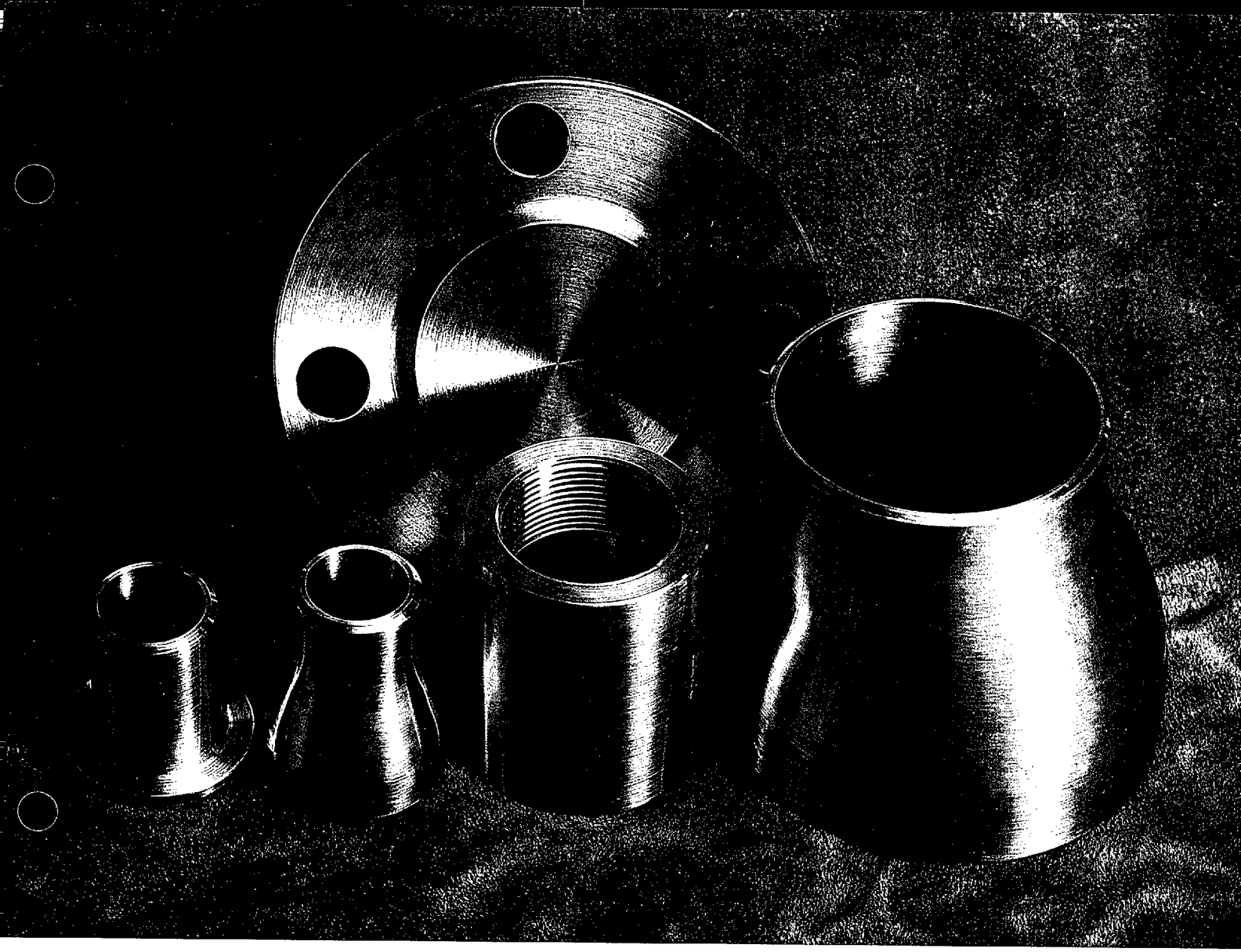
Specimen	Corrosion Rate	
	mpy	mm/y
1	13.5	0.343
2	11.8	0.300
3	15.4	0.391
4	14.6	0.371

[a] Tested according to ASTM G28A.

This heat recovery system at Wisconsin Tissue, Menasha, Wisconsin, USA, incorporates a plate type heat exchanger made, in INCOLOY alloy 25-6MO, by Dairy Equipment Company, a division of DEC International, Madison, Wisconsin. INCOLOY alloy 25-6MO was specified for this application to combat chloride pitting which had lead to the failure of the original AISI 316.



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Machined fittings for the process industries in INCOLOY alloy 25-6MO.

General Corrosion Resistance

Sea Water

In sea water at high flow rates, impingement and cavitation effects can become critical. INCOLOY alloy 25-6MO showed no evidence of any corrosion from ambient-temperature, filtered sea water impacting on its surface at 50 ft/sec (15.2 m/sec).

These characteristics are very important when considering materials for sea water piping on offshore platforms. In the case where chlorination of the sea water is necessary, test results have shown INCOLOY alloy 25-6MO to resist corrosion at levels up to 1.0 ppm of chlorine at 95°F (35°C). At higher chlorine levels, or at elevated temperatures, some corrosion attack can occur, especially in flanged areas or at crevices.

Service equipment for offshore oil and gas platforms must have the same degree of quality in design, fabrication, and material selection as the production equipment itself. In most cases, the severe operating environment imposes a comparable range of demands. INCOLOY alloy 25-6MO provides the reliability needed in service equipment ranging from fire-control systems to desalination units.

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General Corrosion Resistance (continued)

Process Environments

Extensive electrochemical studies and immersion data have been obtained in various salt solutions, especially under evaporative conditions. (See Table 6.) In saturated sodium chloride environments and pH values of 6 to 8, INCOLOY alloy 25-6MO exhibited a corrosion rate of less than 1 mpy (0.025 mm/y). Even under more aggressive oxidizing conditions involving sodium chlorate, INCOLOY alloy 25-6MO maintained a corrosion rate of less than 1 mpy (0.025 mm/y) and showed no pitting even at temperatures up to boiling. (See Table 7.) As the acidity of these brine liquors increases, INCOLOY alloy 25-6MO would offer an excellent choice for crystallizer and evaporator applications. INCOLOY alloy 25-6MO has been used for salt evaporator bodies and shell-and-tube heat exchangers in major chemical facilities.

INCOLOY alloy 25-6MO offers resistance to sulfuric and hydrochloric acids, see Figures 3 and 4.

In the pulp and paper industry, the 6% molybdenum alloys have been extensively used in bleaching environments, especially where aggressive chlorine dioxide reagents are used. Applications include bleach washers, drums, and pipe lines. The chlorine dioxide used for bleaching of kraft pulps is a powerful oxidizer that has been shown to attack austenitic stainless steels.

INCOLOY alloy 25-6MO is also useful to handle corrosive effluents, falling between the epoxy- and brick-lined steel construction and the highly alloyed materials such as INCONEL alloys 625, 686 and C-276 which are required for the very aggressive flue gas environments. INCOLOY alloy 25-6MO offers a cost-effective material for stack liners and outlet ducts where acid condensates can form and cause pitting and crevice corrosion.

Table 6 – Results of a 2-Week Immersion Test in an Evaporated Salt Solution. Test Liquor from an Evaporation Test of Ground/River Waters with Chlorides Plus Sulfides at 250°F (120°C) (17% dissolved solids). Beginning pH 9.8, final pH 7.6.

Alloy	Corrosion Rate		Behavior
	mpy	mm/y	
INCOLOY alloy 25-6MO	<1	<0.025	No localized corrosion.
AISI 316	<1	<0.025	Pitting. Possible SCC at sheared edge.
MONEL alloy 400	1	0.025	Adherent black film. No localized corrosion.

Table 7 – Test Results from a Saturated Sodium Chloride Environment Containing 4% Sodium Chlorate at 145°F (63°C)

Alloy	Electrochemical Prediction*	Immersion Test Corrosion Rate, Behavior	
INCOLOY alloy 25-6MO	Pitting not probable.	<1 mpy (<0.025 mm/y)	No pitting.
MONEL alloy 400	Probable pitting.	<1 mpy (<0.025 mm/y)	Pitted.
INCONEL alloy 625	Pitting not probable.	<1 mpy (<0.025 mm/y)	No pitting.

* Electrochemical estimate of pitting behavior based on polarization curves.

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Table 8 shows the alloy's performance in simulated FGD outlet duct environments. The first series of tests conducted at pH 2.0 and 125°F (52°C) showed accelerated general corrosion of AISI 316, and no attack of INCOLOY alloy 25-6MO. The pH was lowered to 1.5 by the addition of HCl and the temperature was raised to 195°F (91°C). Even in this aggressive environment, INCOLOY alloy 25-6MO was still more resistant to crevice corrosion initiation and propagation than AISI 317LNMo.

Table 8 – Resistance to Crevice Corrosion in Simulated FGD Environments

Alloy	Test Temp °F (°C)	Test Duration (days)	pH	% Crevice Attacked	Maximum Crevice Depth, mils (mm)	Corrosion Rate mpy (mm/y)
First Test Series						
INCOLOY alloy 25-6MO	125 (52)	14	2.0	0	0 (0)	<1, <1 (<0.025, <0.025)
AISI 316	125 (52)	14	2.0	0	0 (0)	34, 31 (0.86, 0.79)
Second Test Series						
INCONEL alloy 625	195 (91)	7	1.5	17, 0	2, 0 (0.05, 0)	<1, 0 (<0.025, 0)
INCOLOY alloy 25-6MO	195 (91)	7	1.5	46, 54	5, 11 (0.13, 0.28)	2, 2 (0.05, 0.05)
AISI 317LMN	195 (91)	7	1.5	71, 96	13, 8 (0.33, 0.2)	5, 6 (0.13, 0.15)

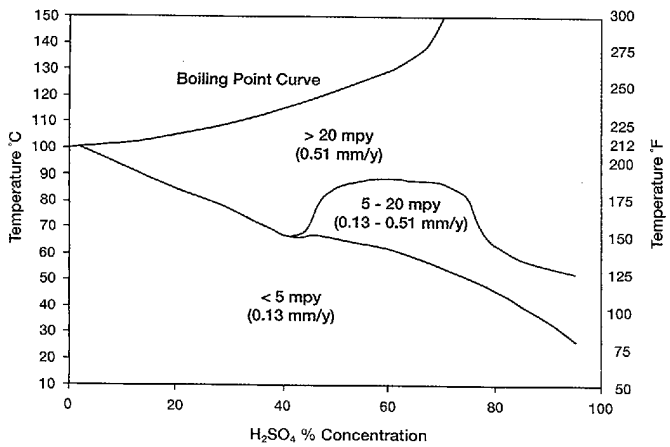


Figure 3 – Isocorrosion chart for INCOLOY alloy 25-6MO in sulfuric acid.

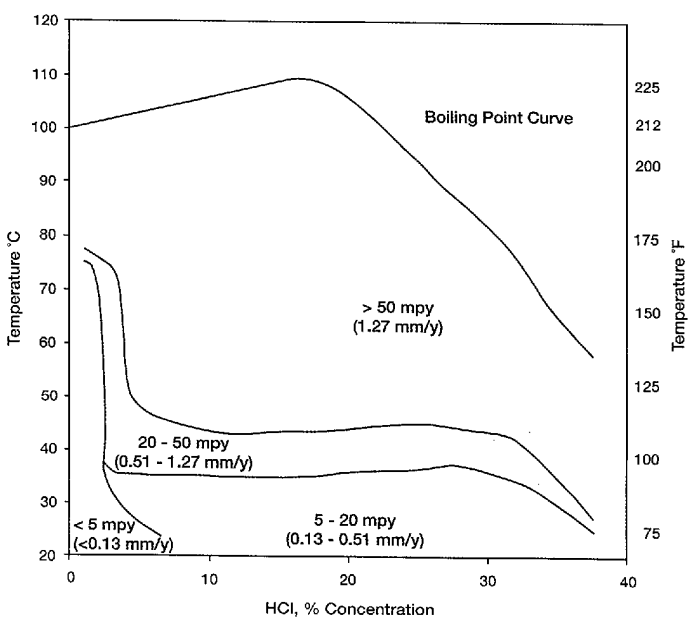


Figure 4 – Isocorrosion chart for INCOLOY alloy 25-6MO in hydrochloric acid.

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General Corrosion Resistance (continued)

Process Environments

INCOLOY alloy 25-6MO has excellent resistance to a variety of acid and mixed acid environments, especially under oxidizing conditions. (See Tables 9 and 10.)

Table 9 – Corrosion Tests in Acid Solutions. One Week Test Duration.

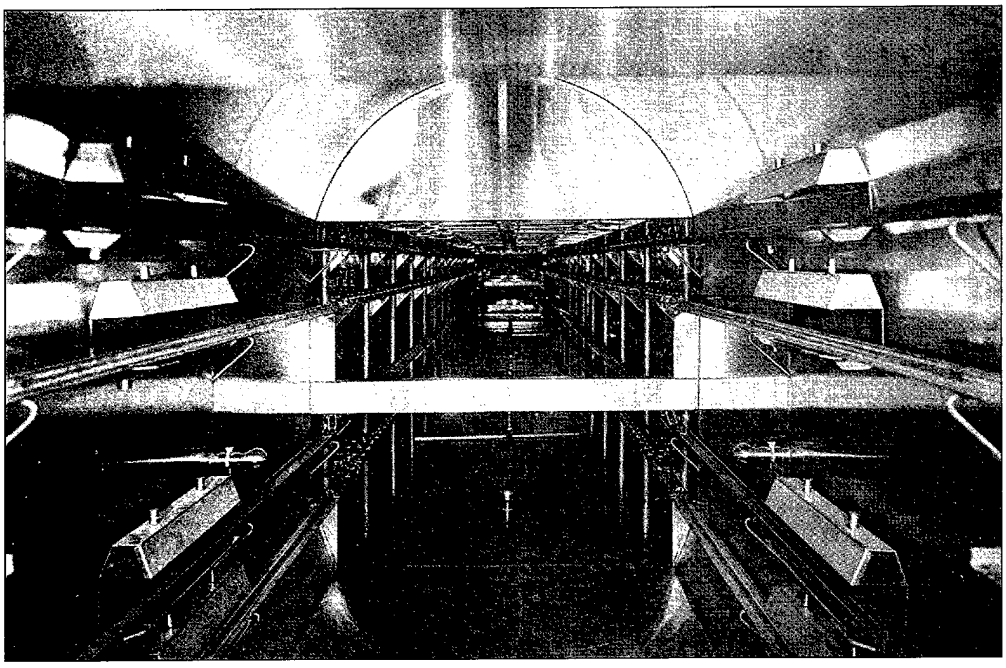
Alloy	Corrosion Rate, mpy (mm/y)			
	10% H_2SO_4 (80°C)	80% H_2SO_4 (50°C)	2% HCl (50°C)	20% HCl* (50°C)
INCOLOY alloy 25-6MO	<1 (<0.025)	<1 (<0.025)	<1 (<0.025)	36 (0.91)
AISI 304	>1300 (>33)	–	33 (0.84)	–
AISI 316	73 (1.85)	61 (1.55)	35 (0.89)	>1300 (>33)

* 24-hour test.

Table 10 – Corrosion Rates for INCOLOY alloy 25-6MO 0.125 in (3.2 mm) Sheet Evaluated in Acid Environments at Varied Exposure Temperatures for 192 Hours

Environment	Temperature °C	Corrosion Rate	
		mpy	mm/y
0.2% HCl	Boiling	<0.10	<0.00
1.0% HCl	90 70	37.00 0.02	0.94 <0.00
5.0% HCl	50 30	43.40 10.60	1.10 0.27
10.0% H_2SO_4	Boiling 70	10.40 2.50	0.26 0.06
60.0% H_2SO_4	70 30	9.40 6.00	0.24 0.15
95.0% H_2SO_4	50 30	18.90 12.50	0.48 0.32
85.0% H_3PO_4	90	0.24	0.01
80.0% $HC_2H_3O_2$	Boiling	<0.10	<0.00

Part of a continuous system for the processing and manufacture of sausages. INCOLOY alloy 25-6MO was specified for its resistance to chloride-ion stress-corrosion cracking and pitting.



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Microbiologically Influenced Corrosion (MIC)

Bacteria recognized as the agents of microbiologically influenced corrosion (MIC) live, thrive and multiply at pits, grain boundaries, weld undercuts, and inside the tube walls of pipes, heat exchanger tubing and other metallic structures associated with water. They come in a number of varieties but fall into two main classifications; anaerobic or reducing-type bacteria (sulfate-reducing and manganese-reducing), and pseudomonas, or oxygen consuming. This latter category is probably the most familiar in that it is a recognizable slime.

MIC can occur in a wide range of applications, especially where stagnant water conditions are present, where pipe systems are not properly flushed, where normally moving water stagnates at plant shutdowns, etc. The standard treatment is the injection of biocides into the water, chiefly chlorine derivatives. But the presence of free chlorine and the means of controlling it are all subject to variables which lessen the effectiveness of the treatment. Operating conditions, water temperatures, the climate of the region where the plant is located, available nutrient levels, available oxygen, evaporation, etc.; all can affect a chemical treatment and lead to a loss of biocide capability.

The susceptibility of the AISI 300 series of stainless steels to MIC has been well documented. One of the current research and testing programs to offer alternative and more resistant materials is focused on the investigation of alloys with a molybdenum content of 6% and higher. The University of Tennessee, Institute of Applied Microbiology carried out laboratory tests and field exposures at a nuclear generating site.[8] The bacteria selected for laboratory tests were:

Pseudomonas Species: Obligate oxygen-consuming bacteria which are also characterized by the profuse quantities of extra-cellular polymers, or "slime", that they produce. These organisms, perhaps the most ubiquitous, were isolated from nodules covering stainless steel weld leaks.

Acinetobacter Species: These bacteria have the capability to oxidize and precipitate ferrous iron in thermodynamically unfavorable conditions.

Bacillus Species: These obligately aerobic (oxygen-consuming) bacteria are characterized by having internal spores that can resist chemical and heat treatments of 212–250°F (100–121°C) for certain periods of time.

Sulfate-Reducing Bacteria (Desulfovibrio): The most cited bacteria involved in corrosion. These obligately anaerobic (cannot grow in the presence of oxygen) bacteria are protected from the oxygen by the growth of other organisms (such as Pseudomonas) on the surface biofilm. Under a mound of other bacteria, the sulfate-reducer not only exists but produces corrosive sulfide anions which accelerate the corrosion process.

Shigella Flexneri Strain A6F: This species, considered pathogenic and not generally used, has been shown to promote very fast crevice corrosion. The organism is characterized by a strong affinity for iron and production of organic acids.

The study concluded that in low total inorganic carbon, low chloride water, a slight increase in chromium, nickel and molybdenum from the 316L composition reduces the risk of MIC attack substantially. Further, iron base alloys, notably the 6% Mo austenitic stainless grades, are more resistant to MIC than 316 stainless steel. Additionally, in field testing of welded specimens in a fire protection system at a nuclear power station, types 304 and 316L stainless steels showed evidence of MIC after only 5 months' exposure. Attack was in the form of rust deposits on the weld, heat affected zone and base metal, caused by iron oxidizing bacteria. Because of its resistance to MIC, INCOLOY alloy 25-6MO is being used in the waste-water piping systems of power plants.

Microbiologically influenced corrosion initiated at a pit on the inside wall of an AISI 304 tube, and propagated from the initial pit in the tube wall.



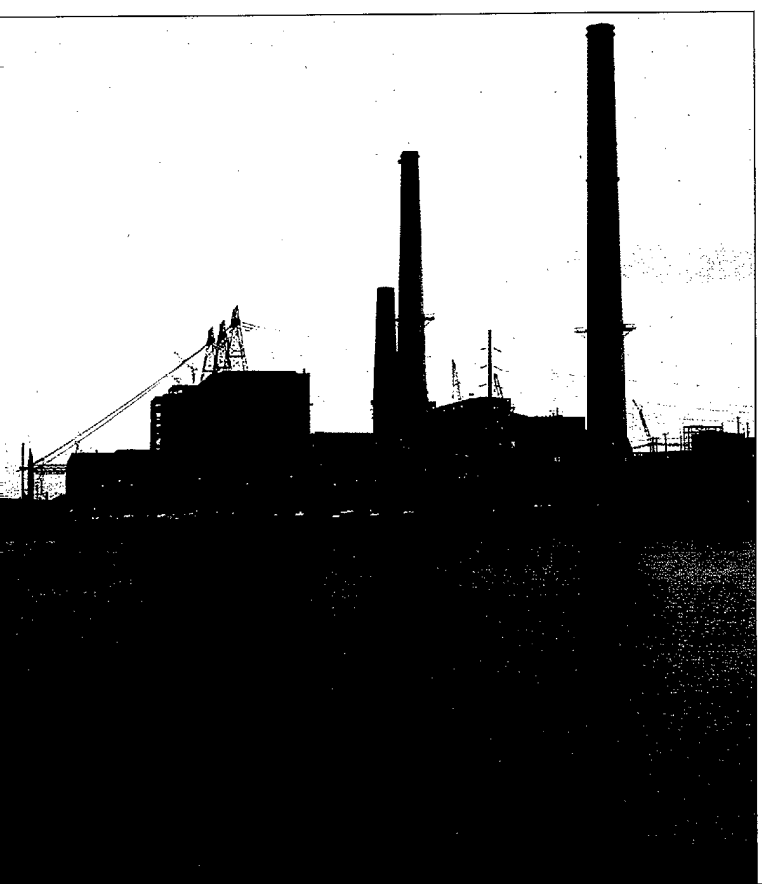
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Mechanical Properties

INCOLOY alloy 25-6MO shows higher mechanical properties than those for the austenitic stainless steels, such as AISI 316L. Table 11 lists typical room-temperature tensile properties of annealed material. The alloy has higher design values than lower strength materials such as 316L, enabling the use of thinner sections.

Table 11 – Room Temperature Tensile Properties	INCOLOY alloy 25-6MO	AISI Type 316	Alloy 904L
Tensile Strength, psi	100,000	75,000	85,000
MPa	690	517	586
Yield Strength (0.2% Offset), psi	48,000	32,000	36,000
MPa	330	221	248
Elongation in 2 in (50 mm) or 4D min,%	42	50	50
Hardness, Rb	86	85	78

Over two million pounds of INCOLOY alloy 25-6MO plate and clad steel plate are in service in power plant pollution control systems such as the flue gas desulfurization equipment at this utility in the U.S.A.



ASME Boiler and Pressure Vessel Code

INCOLOY alloy 25-6MO is approved under the Boiler and Pressure Vessel Code of the American Society of Mechanical Engineers. ASME Code Case 2120 (Ni-Fe-Cr-Mo-Cu-Low Carbon alloy N08926) defines the chemical composition, maximum allowable stress values, and the mechanical property requirements.

ASME Code Case 2120 defines the design stresses up to 750°F (399°C). This design code is applicable for UNS N08926 chemistry requirements *only* (i.e. nitrogen 0.15–0.25%). There is a note attached to this table regarding the use of higher allowable stresses which reads:

"Due to the relatively low yield strength of these materials, these higher stress values were established at temperatures where the short-time tensile properties govern to permit the use of these alloys where slightly greater deformation is acceptable. These higher stress values exceed 2/3, but do not exceed 90% of the yield strength at temperature. Use of these stresses may result in dimensional changes due to permanent strain. These values are not recommended for the flanges of gasketed joints or other applications where slight amounts of distortion can cause leakage or malfunction."

By comparison to 316L, the allowable design stress properties for INCOLOY alloy 25-6MO allow for a significant increase in design stresses, and corresponding benefits in designing with lighter wall or thinner cross-sections.

ASME Section VIII, Division 1 Design Stresses (ksi)		
Temperature °F	UNS N08926[1]	UNS N08926 Conditional
100	26.9	26.9
200	24.1	26.9
300	21.5	26.2
400	19.7	24.8
500	18.7	23.7
600	18.0	22.8
650	17.7	22.4
700	17.5	22.0
750	17.4	21.6

[1]ASME Boiler Code, Table UHA-23

Physical Properties

Some physical properties of INCOLOY alloy 25-6MO are given in Table 12. The values represent annealed material and, except as indicated, are at room temperature. Thermal conductivity, thermal expansion and Young's moduli data are shown in Tables 13, 14 and 15.

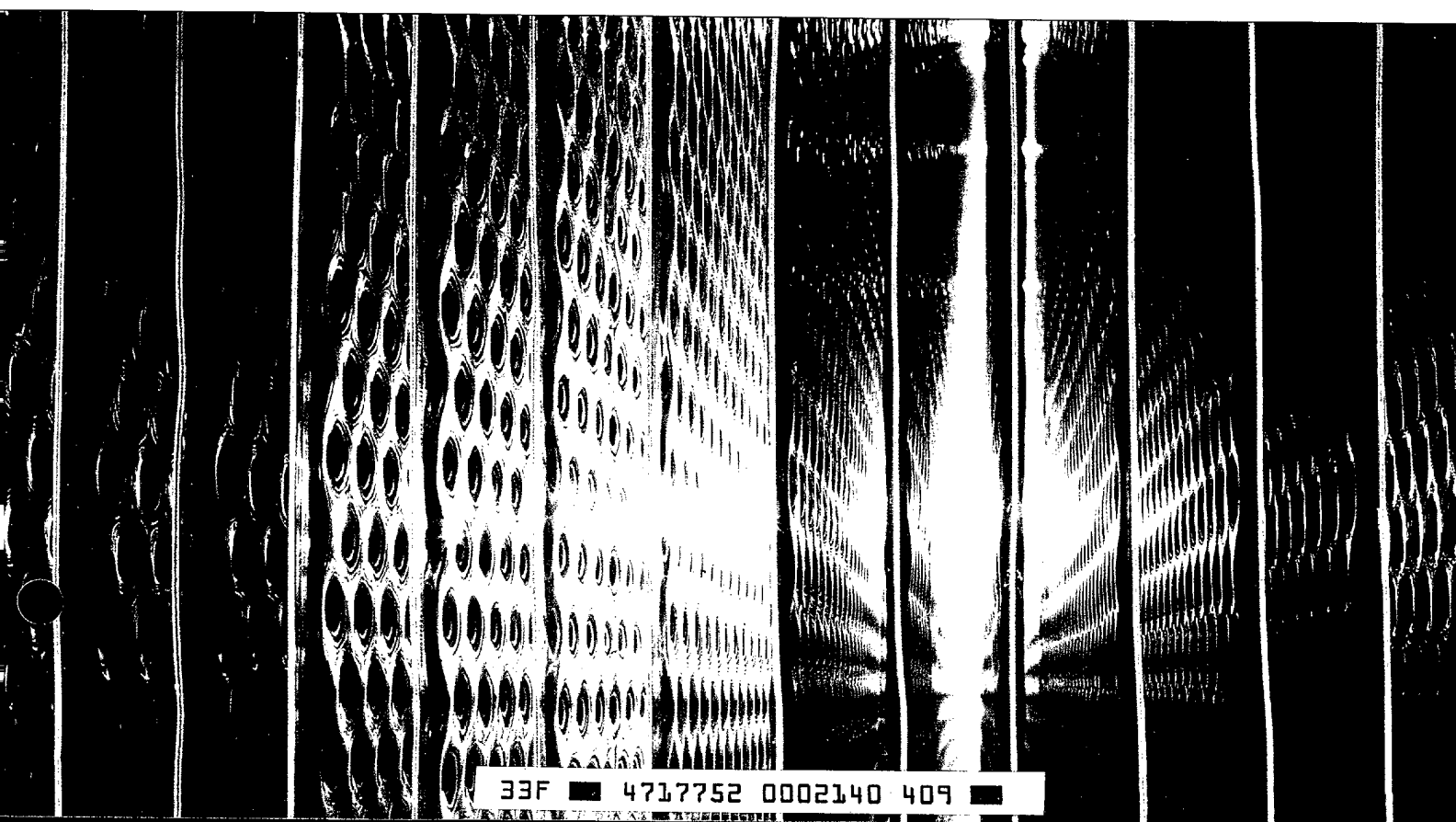
Table 12 – Physical Properties

Density, lb/in^3	0.290
g/cm^3	8.03
Melting Range, °F	2410–2550
°C	1320–1400
Specific Heat, Btu/lb•°F	0.12
J/kg•°C	500
Electrical Resistivity, ohm•circ mil/ft	480
μΩ•m	0.80
Permeability at 200 oersted (15.9 kA/m)	1.005

Table 13 – Thermal Conductivity Calculations

Temperature °C	Conductivity W-cm^{-1} K^{-1}	Temperature °F	Conductivity Btu•in/ft^2•h•°F
23.0	11.5	73.4	79.8
100.0	12.6	212.0	87.3
200.0	14.1	392.0	98.0
300.0	15.9	572.0	110.6
400.0	18.1	752.0	125.5
460.0	19.3	860.0	133.6
500.0	19.4	932.0	134.8
560.0	19.5	1040.0	135.0
600.0	21.9	1112.0	152.0
700.0	24.0	1292.0	166.3
800.0	24.8	1472.0	171.7
900.0	25.2	1652.0	174.7
1000.0	26.3	1832.0	182.5
1100.0	27.9	2012.0	193.8
1200.0	28.5	2192.0	197.8

INCOLOY alloy 25-6MO is used for plate heat-exchangers in the chemical, process and pulp and paper industries.



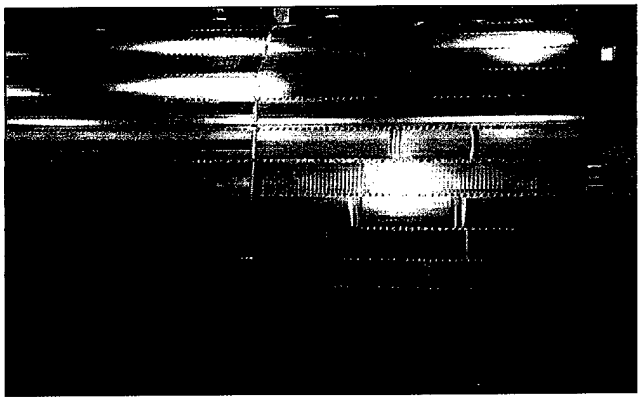
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Physical Properties (continued)

Table 14 – Mean Thermal Expansion Values at Elevated Temperatures for Hot Rolled and Annealed Plate [2100°F (1149°C)/1 h/WQ]

Temperature °F	Coefficient of Expansion in/in/°F x 10^{-6}	Temperature °C	Coefficient of Expansion cm/cm/°C x 10^{-6}
200	8.42	100	15.18
300	8.34	200	15.57
400	8.68	300	16.03
500	8.82	400	16.35
600	8.94	500	16.56
700	9.04	600	16.70
800	9.12	700	17.18
900	9.19		
1000	9.22		
1100	9.26		
1200	9.41		

Reference Temperature = 77°F (25°C).



Many applications for INCOLOY alloy 25-6MO exist in pulp and paper mills, particularly in the bleaching areas where chloride pitting and stress-corrosion cracking are prevalent in stainless steels. Ingersoll-Rand Process Systems, Nashua, New Hampshire, USA, is a major fabricator of bleach plant washer drums, like the one in this picture.

Table 15 – Mean Elevated Temperature Modulus Values for Hot Rolled and Annealed Plate [2100°F (1149°C)/1 h/WQ]

Temperature	Young's Modulus	Shear Modulus	Poisson's Ratio
°F	10^6 psi	10^6 psi	
70	27.27	10.25	0.330
200	26.53	9.97	0.330
300	26.08	9.78	0.333
400	25.59	9.57	0.337
500	24.97	9.37	0.334
600	24.54	9.20	0.332
700	23.92	9.00	0.329
800	23.43	8.79	0.333
900	23.06	8.58	0.344
1000	22.57	8.42	0.340
1100	22.09	8.24	0.340
1200	21.51	7.94	0.354
1300	20.95	7.71	0.359
1400	20.38	7.51	0.357
1500	20.20	7.29	0.385
1600	19.59	7.00	0.399
°C	GPa	GPa	Poisson's Ratio
21	188.0	70.7	0.330
100	182.6	68.6	0.331
200	176.7	66.1	0.337
300	170.0	63.8	0.332
400	163.1	61.3	0.330
500	157.9	58.8	0.343
600	151.8	56.5	0.343
700	144.7	53.3	0.357
800	139.6	50.7	0.377

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Fabrication

Hot working of INCOLOY alloy 25-6MO should be done in the 1800–2100°F (980–1150°C) temperature range. The alloy has good hot malleability, and large rounds or billets are readily forged into components such as valve bodies. Fittings and flanges have been hot pressed from plate stock.

The work hardening response is similar to that of INCOLOY alloy 800 and Type 304 stainless steel. A hardness comparison with other alloy systems is shown in Figure 5 where the hardness values are plotted as a function of the percentage cold work. Optimum ductility can be realized through an anneal at 2150°F (1175°C) although recrystallization of the grain structure can be achieved at lower temperatures.

Annealing has to be done with some caution, since all of the 6% molybdenum alloys have the tendency to suffer from catastrophic oxidation if processed under the wrong conditions. Stagnant oxidizing conditions must be avoided, particularly when heating the alloy above 1700°F (925°C). Attack can also occur at crevice points where flat rolled products are laid on flat surfaces or against one another, thereby resulting in differential furnace atmosphere conditions. Forced gas and air flows in the furnace are recommended. Cooling after annealing should be a rapid air cool or water quench.

Prolonged heating or cooling through the temperature range 950–1560°F (510–850°C) can cause precipitation of undesirable phases such as sigma, or some degree of sensitization, which may lead to intergranular attack or increased susceptibility to stress-corrosion cracking.

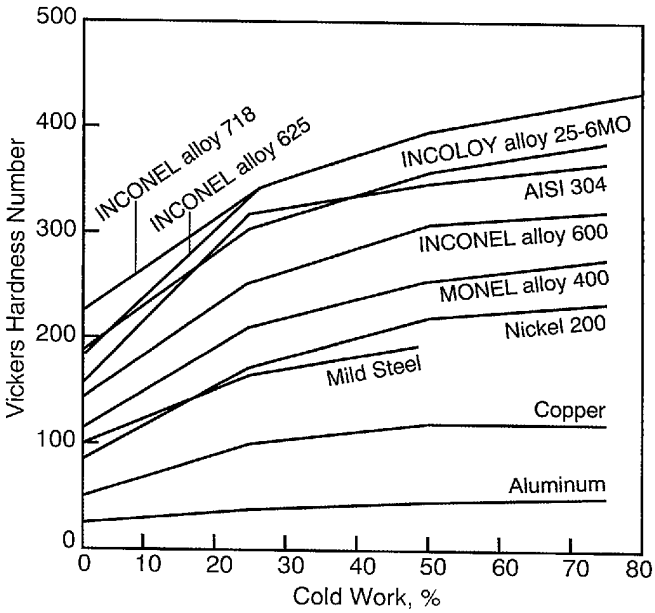
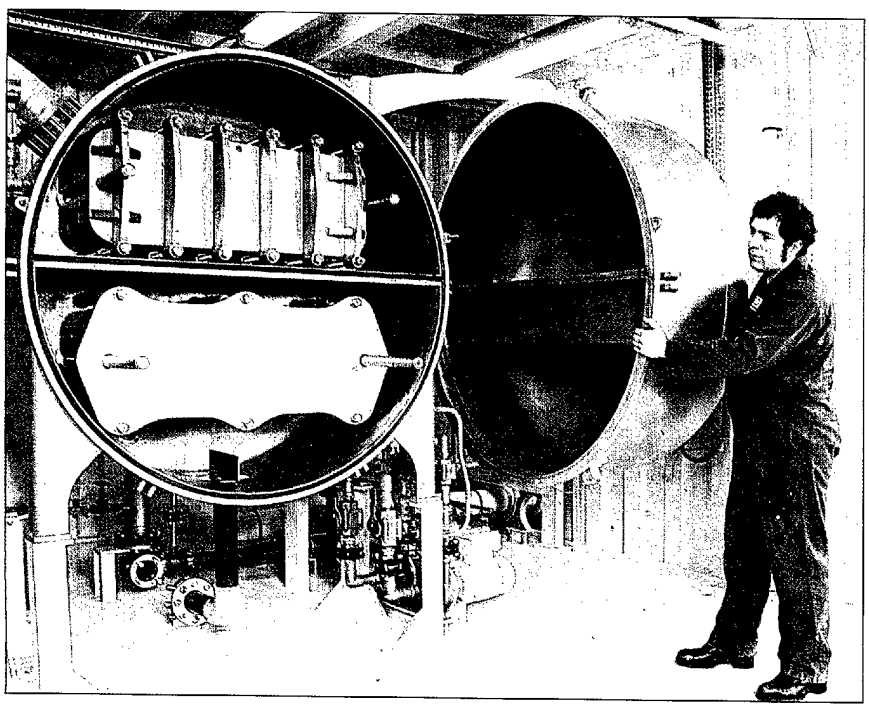


Figure 5 – The work-hardening rate for INCOLOY alloy 25-6MO is similar to that for AISI 304 stainless steel.



INCOLOY alloy 25-6MO was used to fabricate this desalination unit for an offshore platform. The unit was fabricated by KGD Industrial Services Ltd. (Hereford, England) for Alfa Laval Desalt (Copenhagen, Denmark).

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Machining

Machining characteristics of INCOLOY alloy 25-6MO are similar to those of other austenitic alloys. When using carbide insert tooling, rotating at 375 surface feet per minute (114 m/min) with a 0.15 in (0.4 mm) depth of cut, there was no difference in cutting characteristics between 316 stainless steel and INCOLOY alloy 25-6MO. When using high speed steel tooling, 316 stainless steel showed only a 20% improvement in tool life over INCOLOY alloy 25-6MO. This is probably due to the higher nickel content and increased strength of INCOLOY alloy 25-6MO. Consequently, INCOLOY alloy 25-6MO should be machined at slightly lower speeds than those used for stainless steel, using rigid tooling and optimum lubrication and cooling. Positive rake angles should be used with cemented carbide tooling. Cold-drawn, stress-relieved material is always preferred for machining, particularly when the smoothest finish is required. Work hardening characteristics of INCOLOY alloy 25-6MO are comparable to those of 304 stainless steel.

Although drilling tests showed 316 stainless steel easier to drill, INCOLOY alloy 25-6MO produced chips that broke better, allowing better chip control and lubrication. With the appropriate selection of tools, large tube sheets have been readily drilled.

General machining practices are described in the Special Metals publication "Machining" and suggested specific machining parameters are as follows:

High-Speed Steel						
Alloy	Speed		Depth of Cut		Feed	
	SFPM	m/min	in	mm	in/rev	mm/rev
AISI 316L	35	10.7	0.15	3.8	0.018	0.46
INCOLOY alloy 25-6MO	25	7.6	0.15	3.8	0.018	0.46
Carbide						
AISI 316L	375	114	0.15	3.8	0.018	0.46
INCOLOY alloy 25-6MO	350	107	0.15	3.8	0.018	0.46

Drilling with High-Speed Steel Tools				
Alloy	Speed		Feed	
	SFPM	m/min	in/rev	mm/rev
AISI 316L	30	9.1	0.005	0.03
INCOLOY alloy 25-6MO	25	7.6	0.005	0.03



INCOLOY alloy 25-6MO is readily fabricated, for example, to create components like this heat exchanger for the pulp and paper industry.

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Welding

INCOLOY alloy 25-6MO is readily weldable using conventional processes, such as gas tungsten arc welding (GTAW), gas metal arc welding (GMAW), shielded metal arc welding (SMAW), submerged arc welding (SAW), and plasma arc welding (PAW). Past corrosion studies have shown the 6% molybdenum alloys suffer a loss of corrosion resistance when autogenously welded or welded with a matching composition filler metal. Higher molybdenum containing (Mo ≥ 9%), or overmatching filler metals (INCONEL filler metal 625, INCONEL filler metal C-276, or INCO-WELD filler metal 686CPT) should be specified. The equivalent welding electrodes are INCONEL welding electrode 112, INCONEL welding electrode C-276, or INCO-WELD welding electrode 686CPT. INCOLOY alloy 25-6MO can be joined to many types of austenitic or ferritic alloys with these welding products.

All welds should have a slightly larger root opening and included angle between 65° and 75° for V-joints compared to traditional austenitic stainless steels, allowing for improved torch access and lower base metal dilution. When using nickel base welding consumables to weld INCOLOY alloy 25-6MO, the torch should be manipulated aggressively, "puddling" within the weld joint to assure fusion to the base metal and previous weld passes. Ample, but not excessive, filler metal should be used to minimize the amount of base metal dilution, especially for the weld pass which will be exposed to the corrosive medium.

Welding procedures for INCOLOY alloy 25-6MO are significantly less demanding than those for the duplex stainless steels. Unlike the duplex stainless steels, which require a specific heat input (narrow welding parameter) range to maintain proper ferrite number for adequate mechanical and corrosion properties, INCOLOY alloy 25-6MO is fully austenitic, thus allowing for a wider heat input (wide welding parameter) range while maintaining adequate mechanical and corrosion properties. To minimize the amount of segregation which occurs in the fusion zone of the weld deposit, low heat input

Table 16 – Typical Pulsed GMAW Parameters for 0.25 in (6.35 mm) Thicknesses at 120 Pulses Per Second[a]

Weld Pass	Current (amps)	Voltage (volts)	Background Voltage (volts)	Peak Current (amps)	Wire Feed Speed		Travel Speed	
					in/min	(m/min)	in/min	(m/min)
Root	180	25	21	400	430	(10.9)	19.5	(0.50)
Cap	180	25	21	400	430	(10.9)	22.5	(0.57)

[a] For Ni-Cr-Mo type filler metals 0.035 in (0.8 mm) diameter using argon shielding gas at 45 ft^3/h (1.3 m^3/h).

Table 17 – Typical Pulsed GTAW Parameters for 0.25 in (6.35 mm) Thicknesses[b]

Weld Pass	Current (amps)	Voltage (volts)	Wire Feed Speed		Travel Speed		Electrode Extension	
			in/min	(mm/min)	in/min	(mm/min)	in	(mm)
Root	100	9.5	10.0	(254)	2.0	(51)	0.25	(6.4)
2	125	11.0	7.3	(185)	2.9	(74)	0.25	(6.4)
Cap	125	11.0	8.3	(211)	3.2	(81)	0.25	(6.4)

[b] For Ni-Cr-Mo type filler metals 0.035 in (1.6 mm) diameter using argon shielding gas at 45 ft^3/h (1.3 m^3/h).

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Welding (continued)

welding parameters should be incorporated into the welding process when possible to optimize the corrosion resistance of the weld. Inter-pass weld temperatures should not exceed 300°F (150°C) when joining thick sections.

Post-weld heat treatment is not required when overmatching filler metal is used. However, extensive testing has shown that a post-weld heat treatment at 2150°F/5 h/AC (1175°C) can improve the Critical Pitting Temperature (CPT) of the weld by as much as 18°F (10°C) for both autogenous welds and those made with filler metal additions. Pickling after welding, with or without a post-weld solution anneal heat treatment, can further improve the CPT of the welded assembly. A post-weld pickle (nitric/hydrofluoric acid solution or paste[9]) on the corrosive side of the welds for removing the heat tint, can *further* improve the corrosion resistance in bleaching environments. If pickling after welding is not possible, complete shielding on the corrosive side of the weld should be maintained to minimize the amount of heat tint.

For welding processes which require torch shielding gas, 100% argon is recommended, although, 100% helium and mixtures of argon/helium and argon/nitrogen can be used. Backing gases which can be used for shielding the opposing side of the material being welded are argon, helium, nitrogen or mixtures of these gases.

Table 18 – Typical SMAW Parameters [c]

Electrode Diameter in (mm)	Currents (amps)	Voltage (volts)
3/32 (2.4)	40–65	24
1/8 (3.2)	65–90	25
5/32 (4.0)	90–125	26
3/16 (4.8)	125–160	26

[c] For Ni-Cr-Mo type welding electrodes.



INCOLOY alloy 25-6MO was used for the lining of this By-Pass Conditioning Tower (BCT) fabricated by RDM International Corporation, San Antonio, Texas. The tower, in service in a cement manufacturing plant, is approximately 13 ft (3.9 m) diameter, and 85 ft (26 m) high. Working in a process broadly comparable with flue gas desulfurization in coal fired power utilities, INCOLOY alloy 25-6MO has been in service up and downstream from the tower for over three years. It replaced a stainless steel chimney liner which failed early in the aggressive environment.

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Available Products and Specifications

Standard product forms are rounds, flats, plate, sheet, strip, wire, seamless pipe, seamless tube, and forging stock. The products are available in a wide range of sizes. Cold-rolled sheet and strip are produced to one-half ASTM/AMS thickness tolerances as standard practice.

Specifications for INCOLOY alloy 25-6MO (UNS N08926/W. Nr. 1.4529) include:

ASTM (ASME) SPECIFICATIONS

B 625 (SB-625) Plate, Sheet, Strip
B 649 (SB-649) Bar, Wire
B 673 (SB-673) Welded Pipe
B 674 (SB-674) Welded Tube
B 677 (SB-677) Seamless Pipe and Tube
B 472 Billets, Bars for Reforging

ASME CODE CASES (NUCLEAR VESSELS)

N-453	Welded pipe for Class 2 and Class 3 Construction
N-454	Wrought fittings for Class 2 and Class 3 Construction
N-455	Forged flanges and fittings for Class 2 and Class 3 Construction
ASME Code Case 2120 – all product forms	

References

1. "Standard Test Methods for Pitting and Crevice Corrosion Resistance of Stainless Steels and Related Alloys by the Use of Ferric Chloride Solution," Standard G48, *Book of ASTM Standards*, Vol. 3.02, American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.

2. E. L. Hibner, "Modification of Critical Crevice Temperature Test for Nickel Alloys in a Ferric Chloride Environment," *Materials Performance*, March, 1987.

3. J. W. Oldfield, T. S. Lee and R. M. Kain, Institute of Metals Stainless Steel 1984 Conference, Gottenberg, Germany, page 205.

4. E. L. Hibner, E. B. Hinshaw and S. Lamb, "Weld Fabrication of a 6% Molybdenum Alloy to Avoid Corrosion in Bleach Plant Service," presented at TAPPI Engineering Congress, Fall, 1991.

5. H. R. Copson, "Effect of Nickel Content on the Resistance to Stress-Corrosion Cracking of Iron-Nickel-Chromium Alloys," presented at First International Congress on Metallic Corrosion, London, April, 1961.

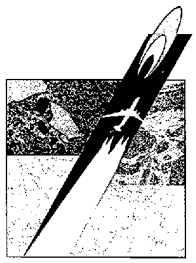
6. Jan Olsson, "Stainless Steels for Harsh Environments," *Materials World*, April, 1993.

7. "Standard Test Methods of Detecting Susceptibility to Intergranular Attack in Wrought, Nickel-Rich, Chromium-Bearing Alloys," Standard G 28 (Method A), *Book of ASTM Standards*, Vol. 3.02, American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.

8. J. R. Crum, "Summary of Microbiologically Influenced Corrosion," Inco Alloys International internal report, February, 1992.

9. Inco Alloys International publication IAI-21, "Fabricating", page 41, formula C.

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INCOLOY® alloy MA956

INCOLOY alloy MA956, (UNS S67956), first developed as an aerospace superalloy, and now used in a range of industrial applications, combines excellent strength and fabricability with outstanding resistance to prolonged exposure up to 1300°C (2400°F). Its exceptional properties result from the mechanical alloying process by which it is made; a process which allows a fine distribution of yttrium oxide particles to be incorporated into a highly corrosion-resistant Fe-Cr-Al alloy.

The alloy has excellent oxidation-resistance up to 1300°C (2400°F) and significant creep strength up to its unusually high melting point of 1482°C (2700°F). Thicker sections may be used up to 1370°C (2500°F). Typical industrial applications include hearth rollers, radiant tubes, furnace muffles, fluidized bed retorts, heat treatment baskets, mesh belts, heat shields, burner nozzles, sensor tubes for thermowells and combustion chamber components for diesel engines.

A full range of both hot and cold-worked mill products is available. The forming and machining characteristics are similar to those of conventional high-chromium ferritic steels and Fe-Cr-Al alloys.

Nominal Chemical Composition, wt.%

Iron	Chromium	Aluminum	Titanium	Carbon	Yttrium Oxide
74	20	4.5	0.5	0.05	0.5

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Physical Properties

Melting point

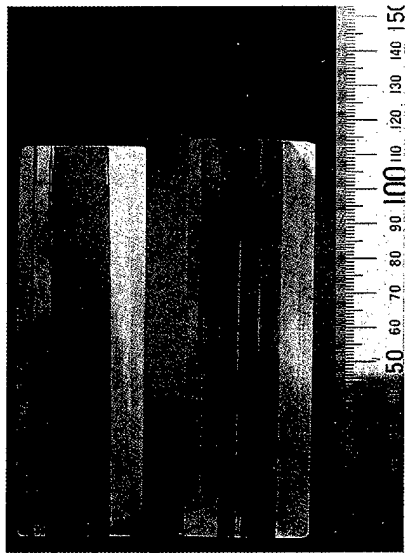
1482°C (2700°F)

The unusually high solidus is 100-200°C higher than for most high-temperature nickel-base superalloys. Even at these extremely high temperatures the dispersion of yttrium oxide provides dimensional and structural stability and prevents embrittlement due to grain growth.

Density

7.2 g/cm³ (0.26 lb/in³)

The low density is only 70% that of molybdenum allowing INCOLOY alloy MA956 to be economically substituted for similar size fixtures in vacuum furnaces.
The lower mass also reduces loading stresses. Unlike molybdenum, the alloy is completely immune to accidental exposure to air at elevated temperatures.



These samples illustrate the elongated grain structure of the mechanically alloyed INCOLOY alloy MA956.

Thermal & Electrical Properties

Temp. °C	Expansion coefficient 10^{-6}·°C	Thermal conductivity W/m·°C	Electrical resistivity μΩ·m	Specific heat J/kg·°C	Young's modulus GPa
20	-	10.9	1.31	469	269
100	11.3	12.2	1.31	491	265
200	11.6	13.9	1.33	519	258
300	11.9	15.4	1.34	547	253
400	12.3	16.9	1.36	575	244
500	12.7	18.4	1.37	602	237
600	13.0	19.8	1.39	630	229
700	13.4	21.2	1.41	658	218
800	13.9	22.6	1.42	686	208
900	14.4	24.1	1.43	714	201
1000	14.9	25.5	1.43	741	191
1100	15.5	27.0	1.44	769	-

Thermal expansion

Compared with nickel-base alloys and austenitic stainless steels, INCOLOY alloy MA956 has the advantage of a relatively low coefficient of thermal expansion.

Thermal conductivity

This property increases with temperature and is higher than for ceramics. This is of particular advantage in radiant tube and heat-exchanger applications.

Resistivity

The material has a high electrical resistance and is suitable for heating applications where the mechanical loading is high.

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Tensile Properties

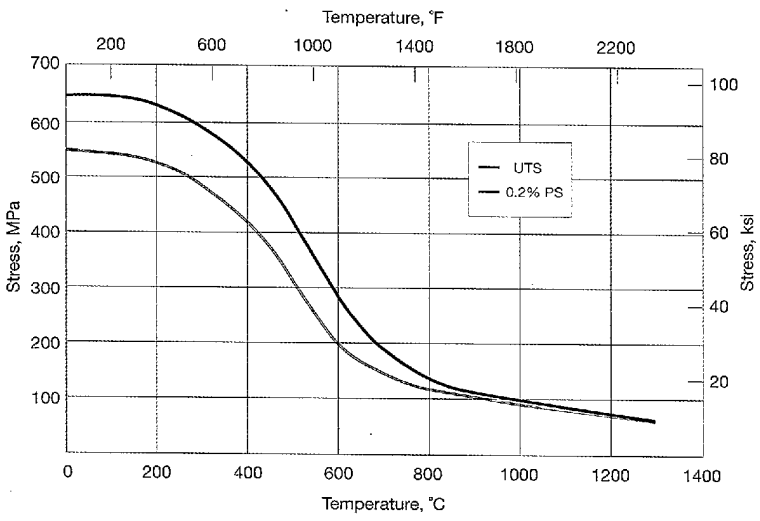
Isotropy

INCOLOY alloy MA956 sheet and plate are rolled in both the longitudinal and transverse directions. After annealing, this results in large pancake shaped grains which minimize directionality in mechanical properties and facilitate the forming of complex shapes from sheet and plate. Tube and bar, mainly processed longitudinally, have a more elongated grain shape and have the highest strength in this direction.

Retention of strength

At the very highest temperatures, the alloy retains significant strength up to the melting point.
At temperatures above 1000°C (1830°F), the alloy is stronger than all conventional, wrought, nickel-base cold-rolled sheet superalloys.

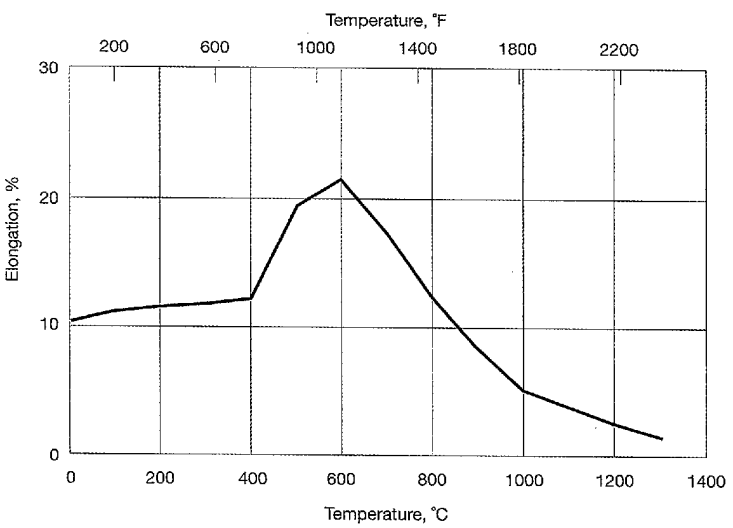
Tensile strength of cold-rolled sheet



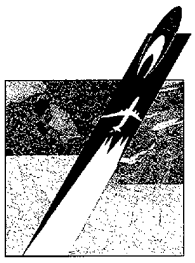
Ductility

Like many ferritic steels in widespread use, INCOLOY alloy MA956 displays a different response to severe deformation below the Ductile-Brittle Transition Temperature (DBTT). The exact point depends on the product form but is generally in the range 0-100°C (32-212°F).
It is recommended that the usual practice is followed for such steels and pre-warming to 150-200°C (300-400°F) is carried out before forming the material. It may be necessary to warm any tools if chilling of the workpiece is likely. Maximum ductility occurs at around 650°C (1200°F) and at this temperature the alloy may be readily formed to complex shapes. At temperatures up to 1300°C (2370°F) there is useful ductility which compares very favorably with alternative ceramic materials.

Tensile elongation of cold-rolled sheet

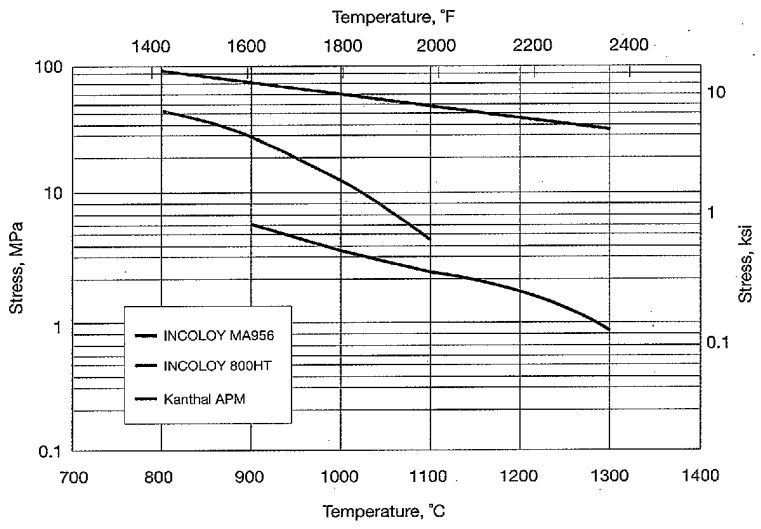


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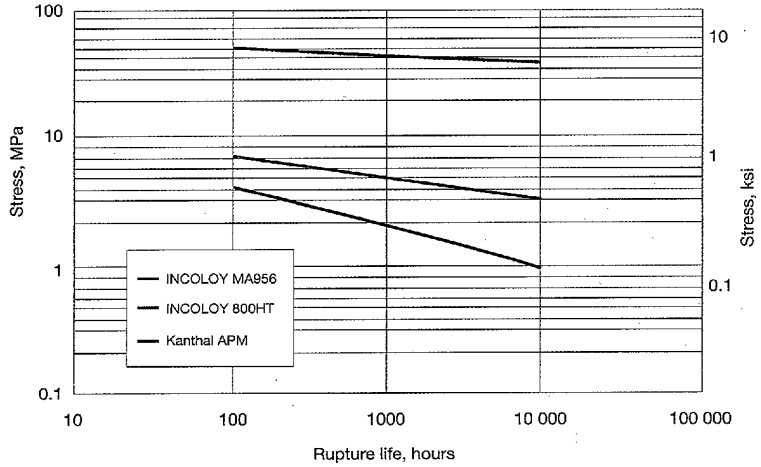
Creep Properties

Comparison of 1000-hour creep-rupture strength



INCOLOY alloy MA956 is made by mechanical alloying. Unlike conventional melted alloys, this permits oxide dispersion strengthening (ODS) with microscopic particles of yttrium oxide. This is effective up to the alloy melting point. In addition to ODS, special processing and heat treatment results in a stable, coarse grain size which further enhances the alloy's high-temperature strength.

Comparison of creep-rupture life at 1100°C (2012°F)



A unique feature of ODS alloys is their extreme long term stability and resistance to creep. The reduction in strength is minimal over very long times compared with conventional Ni-Cr and Fe-Cr-Al alloys, the creep strength advantage above 1000°C (1830°F) being approximately 10-20 times. Using ODS alloys at moderate stresses, it is often possible to disregard the effects of creep.

'APM' is a trademark of Kanthal AB.

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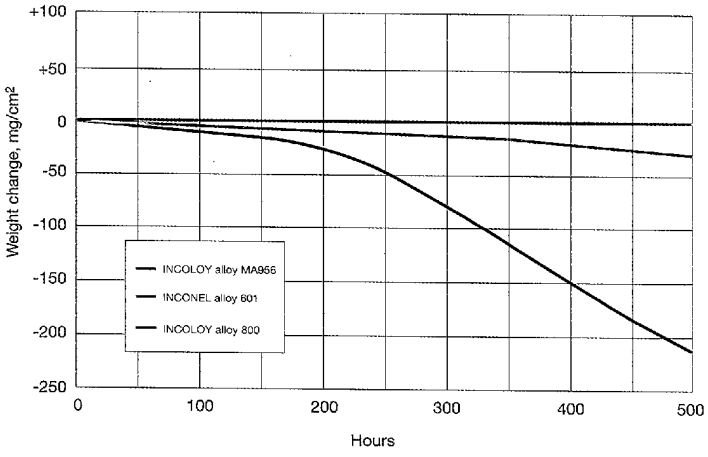
Corrosion Resistance

Oxidation

The high aluminum content of INCOLOY alloy MA956 allows the formation of a thin, highly adherent and protective surface layer of alumina. This process occurs in a wide range of atmospheres, even when levels of oxygen are low. The layer is very effective in reducing the rates of further oxidation. Should the protective layer be mechanically damaged, the high aluminum content of the alloy allows the formation of fresh alumina which heals the affected area. INCOLOY alloy MA956 may be used under oxidizing conditions up to 1300°C (2370°F), much higher than conventional nickel-chromium alloys.

For optimum environmental resistance of machined or cut surfaces, particularly in reducing atmospheres, it is recommended that all traces of lubricants and other foreign matter are removed prior to pre-oxidizing for service in air for 2 hours at 1100°C (2012°F).

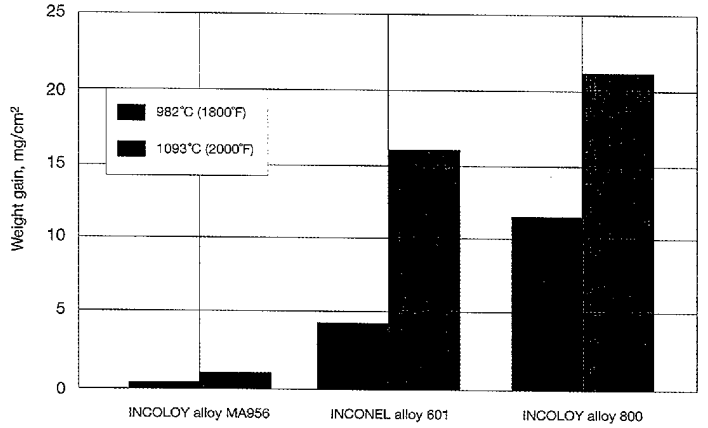
Oxidation resistance at 1205°C (2200°F)



Carburization

The alumina layer is also a barrier to carburization and sulfidation. Carbon diffusion rates into the base metal are greatly reduced, as is the adherence of carbon deposits on the oxide surface. The relatively clean surface under carburizing conditions also maintains excellent heat transfer rates in radiant heating and chemical processing applications, thus reducing the risk of damage to the alloy through overheating.

Comparison of carburization resistance



Test conditions: at 982°C (1800°F). 2% CH_4, 98% H_2 at 1093°C (2000°F). 2% CH_4, 93% H_2, 5% Ar.

Sulfidation

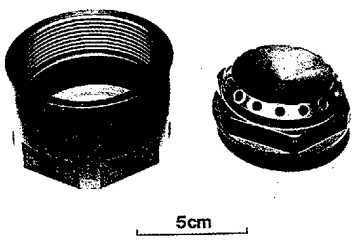
The adherent aluminum oxide scale formed by the alloy provides a surface barrier to sulfur.
The complete absence of nickel in the alloy avoids the formation of harmful nickel sulfide.

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Working

Machining

INCOLOY alloy MA956 is readily machined by all conventional techniques. Its hardness is in the range HV 250-300 (Rc 25-30) and the work hardening rate is relatively low compared with nickel-base alloys. The general machining characteristics are similar to type 410 ferritic stainless steel. Although it is possible to use high-speed tool steels it is usually more economical to employ carbide-tipped tools. Cutting with an abrasive saw requires care to avoid thermal shock which can cause cracking; if possible cooling fluid should be used. If electric discharge machining (EDM) is to be used, the recast surface layer must be subsequently removed by grinding.



An INCOLOY alloy MA956 machined and drilled fuel nozzle.

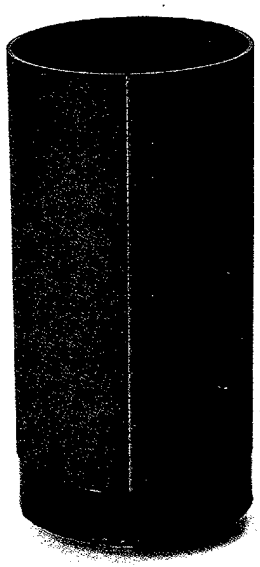
Joining

Conventional TIG welding is possible but produces relatively low strength joints. This process is acceptable for positioning and fillet type welds. Suitable filler wires are INCONEL filler metal 82 for dissimilar metal joints to nickel-base alloys and matching composition wires for welding to austenitic stainless steels. For joining the alloy to itself a Fe-Cr-Al wire is recommended where high-temperature oxidation-resistance is required in the weld. The strongest joints are produced by processes with high energy density such as laser and electron beam welding.
If it is impossible to avoid highly restrained joint designs, post-weld stress relieving should be carried out as soon as practicable to avoid delayed stress cracking. A treatment of 2 hours at 1100°C (2010°F) in air followed by air cooling is suggested. The same cycle may serve as a pre-oxidation treatment provided the surface is cleaned of lubricants and other contaminants.
Brazing, diffusion bonding and transient liquid phase bonding (TLP) are possible if extreme care is first taken to remove the protective

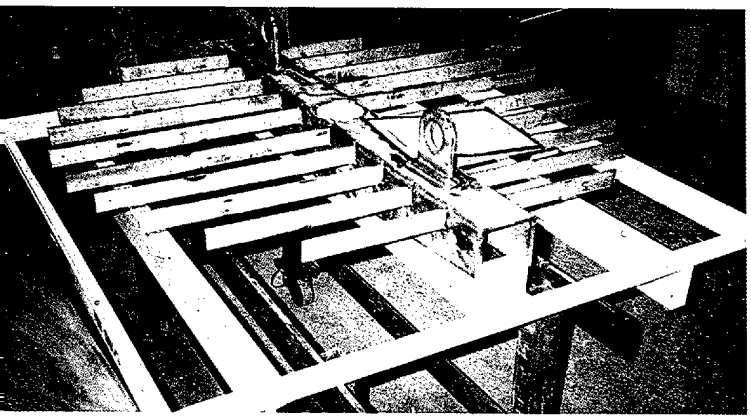
Forming

The nature of INCOLOY alloy MA956 (a ferritic alloy) requires care in high strain rate forming operations such as bending, deep drawing, punching and shearing. For severe deformations it is recommended that the material and tooling is warmed to 150-200°C (300-400°F). alumina film by grinding.
For maximum strength at high temperatures, mechanical joints such as matching composition rivets, pins and threaded connections are often used.

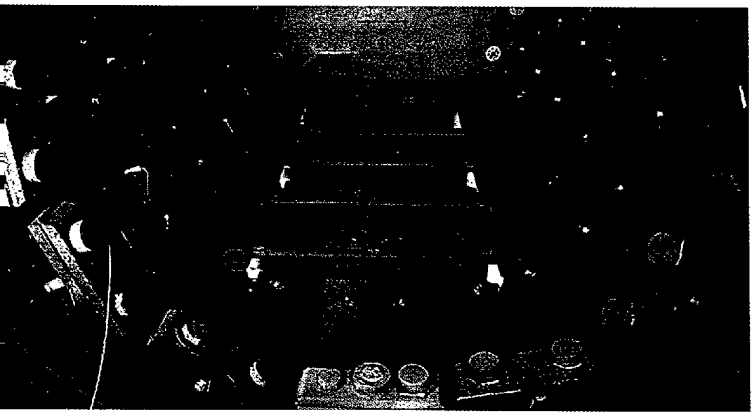
INCOLOY alloy MA956 laser-welded fluidized bed furnace retort cycled between room temperature and 1300°C (2370°F).



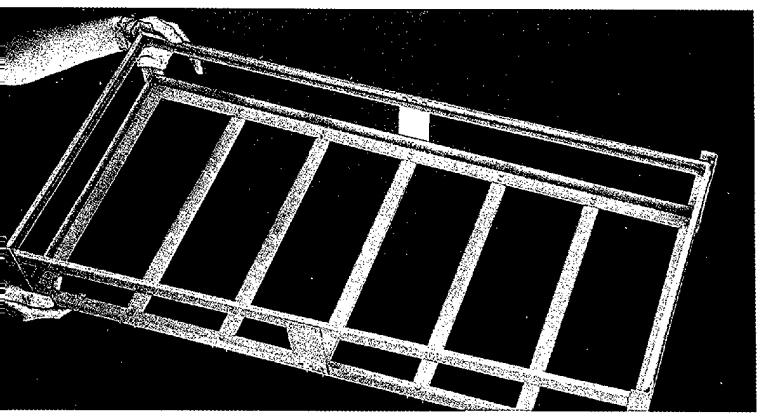
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A vacuum furnace fixture in INCOLOY alloy MA956 to support gas turbine engine components through brazing and annealing operations at up to 2200°F (1204°C).



A vacuum heat-treatment furnace for high-speed steel components in which the removable INCOLOY alloy MA956 hearth has replaced and out-performed the molybdenum original at temperatures up to 1220°C (2230°F).



A heat-treatment tray in INCOLOY alloy MA956 used for pottery firing.

Product Forms

INCOLOY alloy MA956 is available in the following forms and sizes:

	mm	inches
Cold-rolled sheet	0.7-2.0 thick	0.030-0.080
Hot-rolled plate	3-10 thick	0.120-0.400
Tube	10-110 OD	0.400-4.250
Hot-finished round bar	5.5-50 dia.	0.215-2.000
Cold-drawn wire	2.0-8.5 dia.	0.080-0.335
Flat bar from	4 x 11	0.160 x 0.450
to	25 x 75	1.000 x 3.000

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SPECIAL METALS

INCOLOY® alloy 020

Publication number SMC-018

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INCOLOY alloy 020 (UNS N08020) has excellent corrosion resistance in chemical environments containing sulfuric acid, and useful resistance to environments containing chlorides, nitric acid, and phosphoric acid. Other uses include the production of gasoline, solvents, explosives, inorganic and organic chemicals, pharmaceuticals, and food and synthetic materials. INCOLOY alloy 020 is readily fabricated to produce mixing tanks, heat exchangers, process piping, pickling equipment, pumps, valves, fasteners and fittings.

Limiting Chemical Composition, wt %*

Ni[a]	32.0 - 38.0	Nb + Ta	8xC - 1.0
Cr	19.0 - 21.0	Mn	2.0 max.
Cu	3.0 - 4.0	P	0.045 max.
Mo	2.0 - 3.0	S	0.035 max.
Fe	Balance	Si	1.0 max.
C	0.07 max.		

*UNS N08020 [a]Plus Co.

Available Forms

Billet	Rod and/or Bar Products	Hot-Rolled Plate	Sheet and/or Strip	Pipe and/or Tubing	Wire and/or Wire Rod

Specifications

Form	ASTM	ASME	ISO
Bar	B 473	SB-473	9723
Billet & Bar for Re-Forging	B 472	—	—
Spring Wire	B 471	—	—
Wire	B 473	—	9724
Round Weaving Wire	B 475	—	—
Plate, Sheet & Strip	B 463	SB-463	6208
Forgings	B 366	SB-366	9725
Flanges, Fittings & Valves	B 462	SB-462	—
Electric Fusion Welded Pipe	B 474	—	—
Seamless Tube & Pipe	B 729 B 829*	SB-729 SB-829*	6207
Welded Tube	B 468 B 751*	SB-468 SB-751*	—
Welded Pipe	B 464 B 775*	SB-464 SB-775*	—

*General Requirements

Physical Properties

Density		Shear Modulus		Specific Heat		Coefficient of Expansion	
lb/in³	0.291	10^6 psi	11	Btu/lb•°F	0.12	10^{-6} in/in•°F	
g/cm³	8.05	GPa	76	J/kg•°C	500	70-212	8.3
Young's Modulus		**Electrical Resistivity**		**Thermal Conductivity**		70-842	9.4
10^6 psi	28	Ω•circ mil/ft	.651	Btu•in/ft²•h•°F	.85	70-1652	10.0
GPa	193	μΩ•m	1.08	W/m•°C	12.3	μm/m•°C	
						21-100	16.8
						21-450	17.0
						21-900	18.0

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SPECIAL METALS

INCOLOY® alloy 020

Room Temperature Mechanical Properties

Typical Mechanical Properties

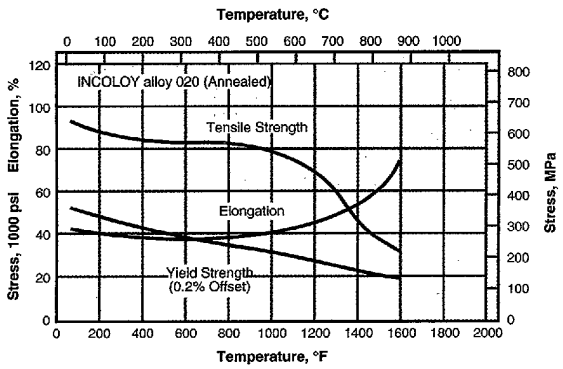
Tensile Strength,	ksi	90
	MPa	620
Yield Strength (0.2% Offset),	ksi	45
	MPa	300
Elongation, %		40

Room Temperature Tensile Test Results of Cold Worked INCOLOY alloy 020

Percent Reduction	Yield Strength (0.2% Offset)		Tensile Strength		Elongation %	Hardness (Rockwell)
	ksi	MPa	ksi	MPa		
0*	44.4	306	95.8	661	38	B83
20	111.0	765	119.8	826	13	C23
30	124.3	857	131.5	907	8	C28
40	127.5	879	135.0	931	7	C27
50	139.5	962	147.5	1017	4	C30
60	139.9	965	149.3	1029	5	C31

*Stabilize Anneal

High Temperature Mechanical Properties



Corrosion Resistance

Sulfuric Acid Corrosion Data for Strip. Duplicate Samples for 168 hours

Temperature		H_2SO_4 %	Corrosion Rate		Average Corrosion Rate	
°C	°F		mpy	mm/a	mpy	mm/a
50	122	70	1	0.025	2	0.051
50	122	70	2	0.051		
60	140	20	0	0	0	0
60	140	20	0	0		
60	140	70	3	0.076	3	0.076
60	140	70	3	0.076		
60	140	90	4	0.102	4	0.102
60	140	90	4	0.102		
75	167	70	5	0.127	5	0.127
75	167	70	5	0.127		
85	185	20	16	0.406	18	0.457
85	185	20	20	0.508		
85	185	70	7	0.178	7	0.178
85	185	70	7	0.178		
85	185	90	15	0.381	15	0.381
85	185	90	14	0.356		
95	203	90	26	0.660	27	0.686
95	203	90	27	0.686		
Boiling		20	20	0.508	17	0.432
Boiling		20	14	0.356		

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SPECIAL METALS

INCOLOY® alloy 020

Welding

Recommended Welding Products

Shielded Metal Arc Welding	Gas Tungsten Arc Welding, Gas Metal Arc Welding
INCONEL welding electrode 112	INCONEL filler metal 625

Heat Treatments

Hot forming should be in the range 1400-2150°F (760-1175°C). INCOLOY alloy 020 is normally used in the annealed condition.

Annealing
1800-1850°F (982-1010°C) / for a time commensurate with section size / AC

Machining

Recommended Conditions for Turning with Single-Point Tools

Hardness	Depth of Cut		High Speed Steel				Tool	Carbide						Tool
			Surface Speed		Feed			Surface Speed				Feed		
								Brazed Tool		Throw Away				
	in	mm	fpm	m/min	ipr	mm/rev		fpm	m/min	fpm	m/min	ipr	m/rev	
75 R_b	0.25	6.35	25-35	7.6-11	0.03	0.76	T-5	150-200	45-60	175-225	55-70	0.02	0.5	C-6
30 R_c	0.05	1.27	50-60	15-18	0.01	0.25	M-36	325-375	100-115	375-425	115-225	0.01	0.2	C-2

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SPECIAL METALS

INCOLOY alloy A-286

INCOLOY® alloy A-286 (UNS S66286/W. Nr. 1.4980) is an iron-nickel-chromium alloy with additions of molybdenum and titanium. It is age-hardenable for high mechanical properties. The alloy maintains good strength and oxidation resistance at temperatures up to about 1300°F (700°C). The limiting chemical composition is given in Table 1. The alloy is austenitic in all metallurgical conditions.

The high strength and excellent fabrication characteristics of INCOLOY alloy A-286 make the alloy useful for various components of aircraft and industrial gas turbines. It is also used for fastener applications in automotive engine and manifold components subject to high levels of heat and stress.

Table 1—Limiting Chemical Composition, %

Nickel	24.0-27.0
Iron	Balance*
Chromium	13.5-16.0
Titanium	1.90-2.35
Molybdenum	1.0-1.5
Vanadium	0.10-0.50
Carbon	0.08 max.
Manganese	2.0 max.
Silicon	1.0 max.
Aluminum	0.35 max.
Sulfur	0.030 max.
Boron	0.0010-0.010

*Reference to the balance of the alloy's composition does not guarantee this is exclusively of the element mentioned but that it predominates and others are present only in minimal quantities.

Publication No. SMC-006

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Data contained in this publication are typical for the properties described, but are not suitable for specifications unless given as limiting.

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INCOLOY alloy A-286

Physical properties

Some physical properties at room temperature are shown along with melting range in Table 2. Values for thermal and electrical properties at high temperatures are listed in Table 3. Physical properties were determined on specimens in the age-hardened condition.

Table 2—Physical Properties

Density,	lb/in³	0.287
	g/cm³	7.94
Melting Range,	°F	2500-2600
	°C	1370-1430
Specific Heat,	Btu/lb •°F	0.100
	J/kg •°C	419
Young's Modulus,	10^6 psi	29.1
	GPa	201
Permeability at 200 oersted (15.9 kA/m)		1.007

Table 3—Thermal and Electrical Properties

Temperature	Coefficient of Expansion[a]	Thermal Conductivity	Electrical Resistivity
°F	10-⁶in/in •°F	Btu in/ft² • h •°F	ohm • circ mil/ft
70	–	88	547
200	9.09	97	–
400	9.16	112	–
600	9.42	126	–
800	9.61	140	–
1000	9.74	155	695
1200	9.91	172	715
1350	–	–	722
1400	9.67	–	
1500	–	–	736
°C	µm/m•°C	W/m•°C	µΩ•m
20	–	12.7	0.910
100	16.4	14.1	–
200	16.5	16.0	–
300	16.9	17.9	–
400	17.2	19.8	–
500	17.5	21.8	–
600	17.7	23.8	1.175
700	17.7	–	1.196
800	–	–	1.217

[a]Mean coefficient of linear expansion between 80°F (27°C) and temperature shown.

Mechanical properties

In the age-hardened condition, INCOLOY alloy A-286 has high strength at room temperature and retains high levels of strength at temperatures to about 1300°F (700°C). Figure 1 shows the effect of temperature on tensile properties.

INCOLOY alloy A-286 has good creep-rupture strength during extended high-temperature exposure. Figure 2 shows stresses for 100 and 1000 hour rupture lives at

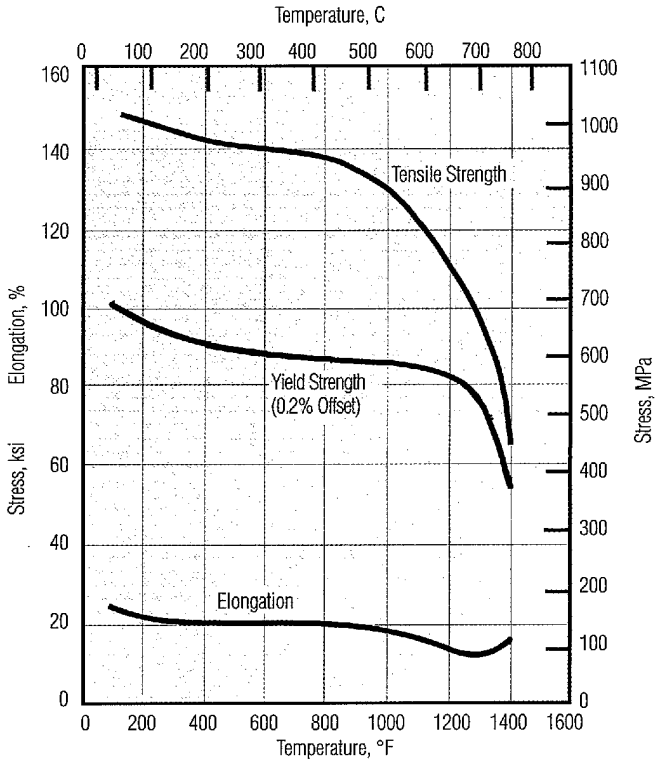


Figure 1. Tensile properties of age-hardened INCOLOY alloy A-286.

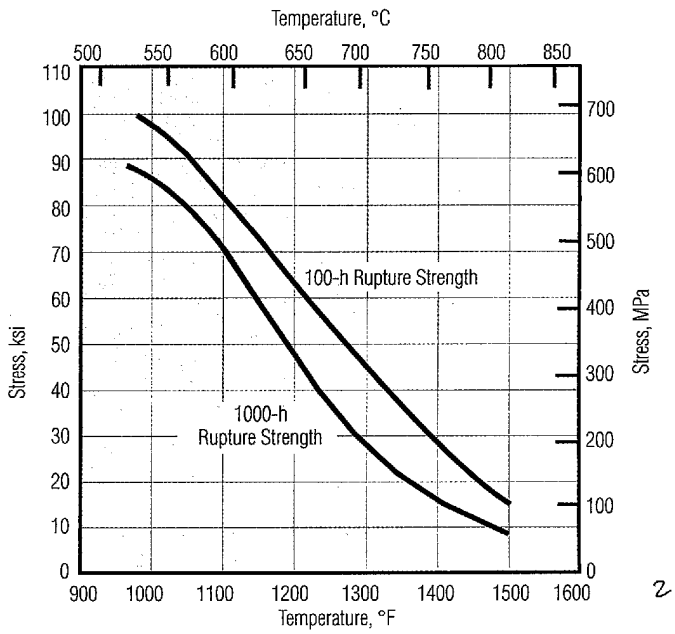


Figure 2. Rupture strength of age-hardened INCOLOY alloy A-286.

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various temperatures. Stress levels to produce creep rates of 1%/100 h and 1%/1000 h are shown in Figure 3.

Mechanical properties reported here are for material given the following heat treatment: solution treatment at 1800°F (980°C) plus age hardening at 1325°F (720°C)/ 16 h and air cooling.

Corrosion resistance

The nickel and chromium contents of INCOLOY alloy A-286 provide resistance to corrosion and oxidation. The alloy has excellent resistance to oxidation at service temperatures up to about 1300°F (700°C).

Fabrication

INCOLOY alloy A-286 is readily fabricated by standard procedures for stainless steels and nickel alloys.

Cold forming should be done on material in the solution-treated condition. Procedures, forces, and work-hardening rates are similar to those for INCONEL alloy 600 and INCOLOY alloy 800.

For hot forming, the metal should be heated to 2100°F (1150°C). Any final reductions at under 1800°F (980°C) should be greater than 10% to avoid formation of large grains during later solution treatment. No forming should be done below 1700°F (930°C).

Machining of INCOLOY alloy A-286 in the solution treated condition should be done using the tooling and procedures given for Group C alloys in the Special Metals publication "Machining." For age-hardened material, the procedures for Group D-2 apply.

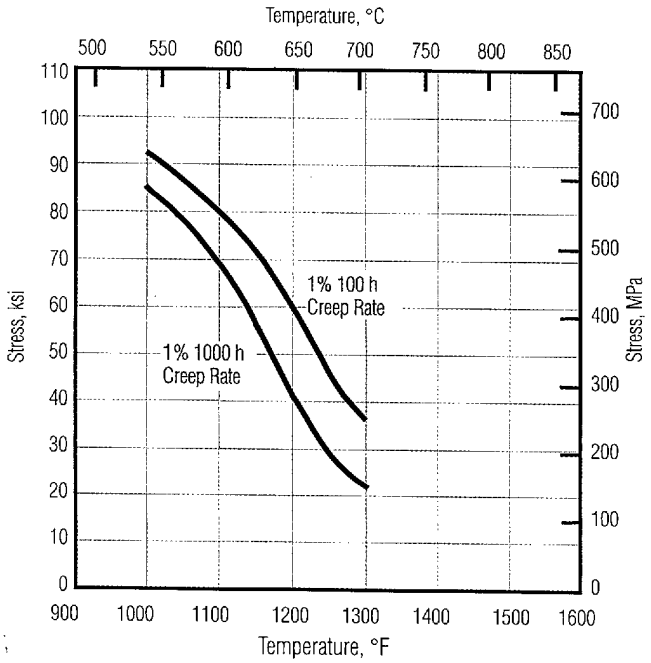


Figure 3. Creep strength of age-hardened INCOLOY alloy A-286.

Heat treatment of INCOLOY alloy A-286 consists of solution treatment at either 1800°F (980°C) or 1650°F (900°C) and rapid cooling followed by age hardening at 1325°F (720°C) for 16 hours and air cooling. The 1800°F (980°C) solution treatment produces the highest creep-rupture strength in age-hardened material whereas the 1650°F (900°C) treatment results in improved ductility and room-temperature tensile strength.

Available products and specifications

Standard product forms are billet, bar, flats, plate, sheet, strip and wire.

Designations and specifications include:

SAE AMS 5525, 5726, 5731, 5732, 5734, 5737, 5858, 5895.
SAE J467.
ASTM A 453, A 638.
ASME SA-638.
BS HR51, HR52, HR650.
AECMA PrEN 2119, 2171, 2173, 2174, 2175, 2303, 2398, 2399, 2417.
AFNOR Z3 NCT 25.

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